



Contemporary issues in trading and microstructure

SMM921: Trading and Microstructure

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May 2, 2026

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$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Dark Pools

modern tool modern problem

What is a dark pool?

- They are **pre-trade opaque** trading venues
 - They do not publish bid/ask quotes
 - They do publish data on executed trades
 - Prices of executed trades are usually linked to quotes observable on **lit** (i.e. not dark) markets
 - For example, trades in a dark pool might be constrained to take place at the mid-quote of a particular lit market
- More than 30 of them in the US alone and almost all US asset managers claim to have used them to execute
- In Europe, over 20 dark venues operate alongside primary exchanges and MTFs

Dark and lit markets comparison

Trader choosing where to submit order to buy/sell stock

Lit market;

Price formation

- Quotes formed from public order prices

Pre-trade, you see;

- Quotes: bids, offers, sizes

Post-trade, you see;

- Trades: prices, sizes

Dark market

Price formation

- Prices constrained by lit market quotes.

Pre-trade, you see;

- No liquidity information at all.

Post-trade, you see

- Trades: prices, sizes

The economics of dark pools

Pros

- **Secrecy:** particularly important for the large/repeat trader
 - No-one knows a dark order is there until it executes
 - Alleviates front-running concerns
- **Competition:** between venues might lead to cheaper outcomes for investors
- **Variety:** variation in market designs might provide better fit to heterogeneous traders

Cons

- **Liquidity:** if liquidity tends to cluster, lack of visibility may reduce liquidity
- **Price discovery:** no independent pricing in the dark; no learning from dark quotes
- **Fragmentation:** fracturing order flow may hamper market quality and makes trading infrastructure more expensive

Dark pools: controversy

Not related to the fundamentals of what dark pools do. Related to lack of honesty in what banks told investors about who and what was allowed in their dark pool.

- **Barclays:** fined \$70mn
 - Claimed to investors that it monitored trading in its dark pool to identify and remove predatory traders
 - In fact, did not do this regularly and sometimes over-rode results of profiling
 - Other traders continued interacting with toxic traders when they thought they were not doing so
- **Credit Suisse:** penalties of \$84mn in total
 - Similar issues with claims that toxic traders were profiled and removed — when in fact they were not
 - Dark pool order information was fed into other Credit Suisse systems
 - Order information was also fed to two HFT firms

Dark pools: current regulation [1]

EU — from double to single volume cap:

- Under MiFID II (2018–2025): a **double volume cap** limited dark trading per stock
 - 4% limit on dark trading on any single dark venue
 - 8% aggregate dark trading limit across all venues
 - Breach $\Rightarrow$ six-month suspension on that stock
- **From October 2025:** the DVC is replaced by a **single volume cap (SVC)** under the MiFIR Review
 - 7% limit on dark trading under the reference price waiver, measured across all EU venues
 - Breach $\Rightarrow$ three-month suspension, EU-wide
 - Important: the SVC covers *only* the reference price waiver — dark trading via negotiated-transaction waivers is no longer capped

Dark pools: current regulation [2]

if you go above 8% there are damages
so you need to measure that every month and ban stocks that go above

UK – a different direction:

- The FCA has **removed the volume cap entirely** post-Brexit, diverging from the EU approach
- Early evidence from the EU SVC: dark volume displaced by caps migrates to **periodic auctions** rather than returning to lit order books – “you are not shrinking dark demand; you are just relocating it”

Empirical evidence on the effects of dark trading [1]

Buti, Rindi and Werner (2011): look at US data covering 11 dark pools in 2009.

- Dark trading is around 5% of volume from this sample and is larger in NYSE rather than Nasdaq stocks
- Dark trading greater in more liquid and larger stocks and in stocks where trading is less volatile
- Accounting for endogeneity:
 - Increased dark trading correlated with lower spreads
 - Increased dark trading associated with lower volatility
 - Increased dark trading may be associated with greater efficiency (result is not unambiguous in the paper)

▶ [Link to journal page for paper](#)

Empirical evidence on the effects of dark trading [2]

Comerton-Forde and Putnins (2015): look at Australian data starting in 2008 for around 500 stocks.

- Dark flow tends to be **less informed** than flow on lit markets
- *High* levels of dark flow lead to increased adverse selection (and spreads) on lit markets
- High levels of dark flow can impede the flow of information into prices
- Important: these effects are **dependent on dark market share**
 - Results on efficiency and price discovery can be *reversed* when dark trading is minor

▶ [Link to JFE published paper](#)

Dark trading: summary

Putting things together:

- Just because trading is **dark** does not mean that it is bad:
 - Low levels of dark trading are innocuous, perhaps even beneficial for market quality
- **But:** some evidence that high levels of dark trading can be damaging to efficiency and lit-market liquidity
- Important to take a **stock-by-stock** perspective (as the economics is felt at the stock not market level)
- The EU's regulatory journey illustrates this complexity:
 - The DVC (2018–2025) could be argued to be blunt and over-aggressive, potentially banning dark trading in stocks where it was not causing harm
 - The new SVC (from Oct 2025) is simpler but still catches only one type of dark waiver
 - Evidence so far suggests caps relocate dark volume to periodic auctions rather than eliminating it
 - The UK's decision to remove caps altogether is the alternative regulatory view

Algo trading and HFT: an introduction

We will now discuss:

- Automated trading systems and their role in modern markets
- Distinguish algo trading from high-frequency trading
 - Algo traders execute client orders
 - HFTs trade for their own account
- What do HFTs do? Helpful or harmful?
- Flash crashes and market robustness
- Recent academic evidence on HFT
- Summary and policy responses

HFT: what are the issues?

Basic research question: has the rise in automation in equity (and other) markets caused those markets to operate less well?

- How do developed equity markets really operate these days?
- How can we measure the quality of operation and the impact of automation on this?
- What are the implications for regulation and market design?

Equity markets: the way the world used to be

Players and places:

- **Ultimate customers:** buyers and sellers of stock for savings, speculation and/or risk-sharing
 - Pension funds and insurance companies trading on our behalf
 - Wealthier individuals wishing to trade stock directly
 - Proprietary trading desks at banks taking positions
- **Broker/dealers:** were middlemen — the customer gave the trade to a broker/dealer who took it to market and executed it
- **The exchange:** traditionally a physical location where dealers came together to trade
 - Sell orders got matched with buy orders (on behalf of customers)
 - The exchange often employed a **market-maker** to smooth trading in stocks
 - Often, exchanges effectively had **monopolies** in organised secondary trading

Equity markets: the way the world looks today

Today: every step of the trading chain may be replaced by a computer.

- **Customers:** real customers (asset managers) still exist
- **Brokers/dealers:** the customer order is fed into a computer, which decides on the optimal way to execute — perhaps chopping it up and sending it to multiple venues
- **The exchange:** no longer a physical location; a **virtual** matching engine where computers interact
 - Designated market makers have all but disappeared for liquid stocks
 - Fragmentation: many competing trading venues
- **New players:** **high-frequency traders (HFT)**
 - Work at hedge funds or proprietary trading firms
 - Speculate on price movements at sub-second frequencies
 - Computers make trading decisions and determine execution strategy
 - Human involvement in individual trading decisions is minimal

Defining HFT

What are the key features of HFTs?

- **Automation:** lack of human intervention in individual quoting and trading decisions
- **Speed:** importance of time-to-market or latency (e.g. through co-location, or microwave links)
- **Traffic:** often very large rates of messaging activity, with orders submitted and then cancelled milliseconds later
- **Note:** regulators do (and should) distinguish HFT activity from **algorithmic** activity which seeks to execute **client** flows in an agency capacity

Algo trading [1]

What differentiates algo trading from HFT?

- Both employ fast computer systems to place orders and execute trades
- An algo trader is typically executing an order for a **client** — the inventory acquired or delivered does not affect the trader's own position
- An HFT is trading for his/her own book — the trading is **proprietary** — so any completed trades enter the HFT's inventory

Another way of thinking about this:

- **Algo trading:** the trade construction is done outside the system and the algo trader focuses on execution
- **HFT:** trade construction and execution are part of the same integrated system

Algo trading [2]

Algo trading example: a client passes you an order to buy £1mn of MKS.L which is currently trading at a price of 348 GBX.

- Trade equates to just over 287,000 shares, a significant chunk of daily volume
- Client instruction: buy the stock across the day using a **VWAP benchmark**
 - Execution judged by comparing average price paid with the day's VWAP
- **Execution strategy:**
 - Divide the trading day into 34 fifteen-minute intervals and split the parent order into 34 child trades
 - Size each child trade by: parent order size $\times$ average fraction of daily volume in that 15-minute interval
 - If 5% of a day's volume typically executes in the first 15 minutes, the first child order should be 5% of parent order size
 - Orders in each window may be passive or aggressive, but the target fill should be met in each interval
 - This strategy should get you close to the VWAP

Algo trading [3]

Some common algorithmic execution strategies (based on the UBS AES / Advanced Execution Services platform, formerly Credit Suisse AES):

- **Sniper:** never shows bids or offers; trades aggressively up to required size
- **Inline:** uses an arrival-price benchmark and tries to minimise implementation shortfall versus that benchmark
- **VWAP:** uses a historical volume curve to trade relative to a VWAP benchmark (as in the previous example)
- **TWAP:** sends orders into the market regularly through time
- **Pathfinder:** routes orders to all available liquidity venues to minimise costs

These are algorithms to *execute* pre-defined client orders. An HFT system would integrate trade construction and execution into a single system.

What do HFTs do? [1]

- **Market making:** providing liquidity in financial markets; supplying immediacy
 - May allow the realisation of gains from trade and thus be economically valuable (Menkveld and Jovanovic, 2016)
 - May reduce bid-offer spreads and increase market depth
 - Market making is risky and valuable, so HFTs should be rewarded for doing it
 - Note: this service is *without obligation*, unlike traditional designated market makers
- **Statistical arbitrage:** speculating on the prices of related securities and attempting to profit from mis-pricings
 - Not a new activity, but a sped-up version of an older practice
 - Signals employed may have changed (e.g. order-book imbalances) and scope may have changed with computerisation and fragmentation

What do HFTs do? [2]

- **Market manipulation and market abuse?** What newspapers focus on when they write about HFT.
 - **Layering:** attempt to push prices in one direction by adding liquidity to one side and making the order book lop-sided – then trading against the artificially moved price
 - **Quote-stuffing:** send a stream of orders that are cancelled almost instantly, perhaps to overload others' systems and slow down market data feeds
 - **Momentum ignition:** exploit predictable counter-parties by trading and pushing prices away from them, then closing your position at a profit
- All three practices are illegal in the EU (Market Abuse Regulation) and the US (SEC market manipulation rules)
- Enforcement has increased: several firms have been fined for layering and momentum ignition strategies

Latency arbitrage: an example [1]

Take a European setting.

- The primary market is the LSE and we are looking at trade in BARC.L
- A dark pool also runs a market in BARC.L where orders are matched at the mid-quote from the primary exchange
- **But:** to do this, the dark pool has to take a price feed from the LSE
 - Simple physics tells us that the mid-quote data the dark pool has will be slightly **delayed** relative to the LSE mid-quote — call this delay X
 - It takes time to transmit information from the LSE server to the dark pool server: e.g. 320×10^{-6} seconds between London and Slough (where BXE is located)

Latency arbitrage: an example [2]

- **Latency arb:** assume you are an HFT firm
 - Your software and links are such that you can receive mid-quote data from the LSE and trade on the dark pool in less than X
 - So you can trade at **stale prices** if you want to
 - If the LSE mid-quote has risen (fallen) a lot, buy (sell) on the dark pool at a stale, low (high) price and then sell (buy) at a profit on the LSE

Lesson: this shows that simple technological edges and complexity in market structure can create opportunities for fast traders to exploit slow traders – even without any private information advantage.

So what? The key question [1]

So why should regulators and investors care?

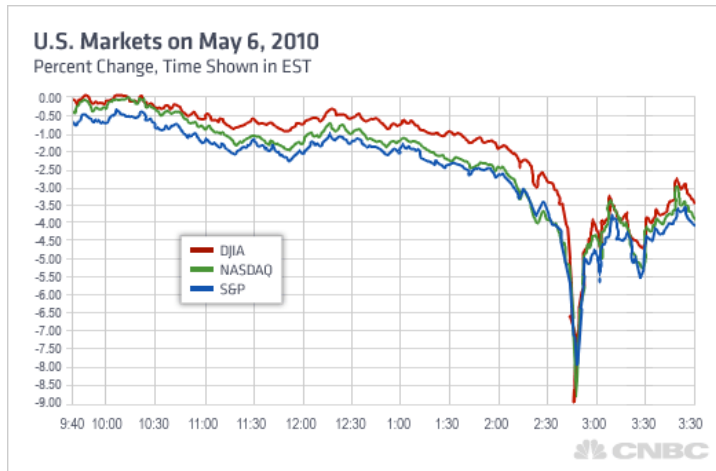
- **Speed differentials** may facilitate HFTs abusing other traders (latency arbitrage as shown above)
- **Control failures** in algorithmic systems can cause severe market disruption very quickly – see the Citigroup incident (slide 25): \$444bn of erroneous orders, \$1.4bn executed, European indices down 4%, in under 15 minutes
- **Complex interactions** of algorithms are hard to model and the resulting outcomes are hard to understand or anticipate
- If HFT **increases investor trading costs**, rational investors may invest less in information production $\Rightarrow$ markets become less efficient
- An **arms race** in fast links and tech is socially unproductive and the cost of the race falls on all market participants, not just HFTs (Aquilina, Budish and O'Neill, 2022)

So what? The key question [2]

What questions should we ask as academics in economics/finance?

- Has computer-based trading made market prices less efficient or less liquid on average?
- In computer-based markets, is it more expensive for final customers (pension funds, retail investors) to buy and sell stock?
- Has reliance of markets on computers made markets more volatile and more prone to violent swings?
- Is the arms race in speed socially costly and, if so, how should market design be reformed to address it?

Evidence 1: The US flash crash — May 6th, 2010



This event has been interpreted by some as clear evidence that computer-based trading has led to destabilised markets. Kirilenko et al. (2017, JFE) provides some academic analysis.

Evidence 2: the Citigroup incident – 2 May 2022 [1]

What happened?

- A trader at Citigroup's London desk intended to sell a basket of equities worth **\$58 million**
- He entered the desired notional into the wrong field in the order management system, creating a basket worth **\$444 billion**
- Automated controls blocked \$255bn – but **\$189bn** was routed to Citigroup's VWAP algorithm
- **\$1.4bn** of sell orders executed across European exchanges before the trader cancelled the remaining orders at 09:10am – only 22 minutes after entering the order

Evidence 2: the Citigroup incident – 2 May 2022 [2]

Market impact and regulatory response:

- European stock indices fell **more than 4%** within minutes before recovering – a flash crash caused by a typo
- FCA fine: £27.7mn; PRA fine: £33.9mn (£61.6mn total, May 2024)
- FCA finding: *“absence of certain preventive hard blocks and inappropriate calibration of other controls”*
- The trader was able to override hundreds of warning pop-ups with a single click, indicating that tech design can be part of the problem

Literature review: theoretical work

Theoretical work favourable to CBT

- *Martinez and Rosu (2011)*: CBT traders exploit a speed advantage as aggressive traders; information is therefore more quickly reflected in prices
- *Menkveld and Jovanovic (2016)*: CBT market makers can reduce adverse selection because they react quickly to news; can increase social welfare where adverse selection was already a problem

Theoretical work unfavourable to CBT

- *Cartea and Penalva (2011)*: parasitic CBT trades between retail traders and professional dealers, raising retail trading costs
- *Li (2017)*: HFTs with a speed advantage can front-run slow traders whose order placements are predictable — buying ahead of them and selling back at a higher price
- *Budish, Cramton and Shim (2015)*: the **design** of continuous limit order books creates arms-race rents; switching to frequent batch auctions would eliminate them

Academic empirical work: measuring market quality

First task: some measures of how well stock markets are doing their jobs.

- **Execution costs:** how much does it cost to do a trade for a customer?
- **Information content of trades:**
 - What is the information content of computer-based trades versus other trades?
 - Do computer-based trades contribute significantly to **price discovery**?
- **Efficiency:** is computer-based trading associated with prices reflecting information more quickly and more accurately?
- **Robustness:** is CBT associated with more fragile markets — larger tail events, flash crashes?

Empirical evidence: positive effects of HFT [1]

Hendershott, Jones and Menkveld (2011): [▶ Paper](#)

- Exploit automated quote dissemination on NYSE as instrument
- Algo trading reduces spreads and adverse selection

Brogaard, Hendershott and Riordan (2014): [▶ Paper](#)

- Nasdaq data with HFT firms flagged; HFT trades and permanent price changes are positively related — HFTs improve price discovery

Brogaard, Hagstromer, Norden and Riordan (2015): [▶ Paper](#)

- Speed upgrade at Nasdaq OMX Stockholm causes large liquidity improvement — spreads narrow and depth rises

Empirical evidence: positive effects of HFT [2]

Chakrabarty, Moulton and Wang (2022): [▶ Paper](#)

- HFTs reduce price inefficiencies after *low-attention* earnings reports by 65–100%
- Machines compensate for human attention constraints

Overall: strong evidence that HFT market-making reduces average trading costs. Evidence on price discovery is positive but concentrated in liquid, large-cap stocks.

Empirical evidence: negative effects of HFT [1]

Shkilko and Sokolov (2020): [▶ Paper](#)

- The fastest HFTs use microwave links; slower traders use fibre-optic cable — rainfall disrupts microwaves, eliminating the speed advantage temporarily
- When it rains: trading costs fall, adverse selection drops, volatility declines
- Interpretation: speed differentials allow fast traders to pick off slow traders, damaging liquidity

Korajczyk and Murphy (2019): [▶ Paper](#)

- A Canadian regulatory change reduced HFT order submission rates — show that HFT raises institutional trading costs
- Some cost comes through *back-running*: HFTs can infer the trading intentions of slow traders who use predictable algorithmic execution patterns

Empirical evidence: negative effects of HFT [2]

Quantifying the arms race: Aquilina, Budish and O'Neill (2022): ► Published paper (QJE 2022)

- Uses message data (not trade data) which includes *failed* attempts to trade or cancel, allowing researchers to see winners **and** losers of a speed race, not just the winner

Key findings (using FTSE 100 and FTSE 250 data):

- **Latency-arbitrage races were very frequent:** approximately **one race per minute per stock** for FTSE 100 stocks
- **Races were extremely fast:** the modal race lasts **5–10 microseconds**
- **Races created substantial volume:** about **20%** of total volume in FTSE 350 stocks related to latency-arbitrage
- **Trade-level cost:** authors argue that latency arbitrage generated a cost equivalent to a 0.5 b.p. tax on trading.
- **Total cost:** authors argue that races may generate around **\$5 billion per year** globally in rents extracted by fast traders from slower market participants
- **Concentration:** the top 6 firms generate 80% of wins and losses

Empirical evidence: negative effects of HFT [3]

Overall: emerging consensus that *aggressive* HFT strategies can damage liquidity and extraction of rent from slow traders through arms-races is quantitatively significant.

Finally: policies to restrict HFT and AT? [1]

Many have been suggested and some tried:

- Minimum resting times for orders
- Financial transactions taxes
- Maximum order-to-trade ratios
- Regulatory access to code of algorithmic trading engines
- Redesign of order priority rules in markets
 - E.g. frequent batch auctions to replace continuous trading (Budish, Cramton and Shim, 2015)
- Better-designed circuit breakers in markets

Finally: policies to restrict HFT and AT? [2]

Where things stand:

- Minimum resting times have been implemented on EBS and Reuters (FX), reportedly with little effect on liquidity
- The SEC proposed a major US equity market structure reform package in 2022–23 (Order Competition Rule, Best Execution Rule) which would have required retail orders to be exposed to open auctions before execution
- The reform package was **withdrawn in 2025** under the incoming SEC Chair, as part of a broader deregulatory push
- EU: MiFIR Review in force from March 2024 addresses some structural issues but does not target HFT directly

HFT: Summary [1]

Tentative conclusions:

- **On balance:** HFT market-making is broadly positive for average market quality, generating tighter spreads and faster price discovery. But the consensus has shifted and is becoming less uniformly positive.
 - HFT market-maker trades generate lower costs for customers in normal conditions
 - Evidence on price discovery is positive but concentrated in large-cap, liquid stocks
 - Emerging evidence (Aquilina et al., 2022) shows that the arms race in speed imposes significant costs on other market participants
- **Some HFT strategies are clearly harmful:** aggressive latency arbitrage picks off slow traders; back-running exploits predictable algorithmic execution
- **Data constraints** limit progress: HFT flagged datasets remain rare and expensive; most evidence is indirect or from specific regulatory shocks

HFT: Summary [2]

- There is clear evidence of markets experiencing periodic **fragility** (flash crashes, algo failures as with Citigroup 2022)
 - Perhaps restricting HFT would attract liquidity suppliers with longer horizons, making markets more stable
 - But by some measures HFTs are the most efficient liquidity providers in normal conditions
- Specific market-design interventions to alleviate robustness problems may have unintended consequences
- The **policy frontier** is now about:
 - Whether frequent batch auctions can replace continuous trading without sacrificing liquidity
 - How to regulate the arms race without eliminating legitimate competition
 - Whether current disclosure requirements are adequate for regulators to monitor HFT strategies in real time

The retail trading revolution

In recent years we have seen a structural shift in who trades and how they trade:

- **Zero-commission trading** (Robinhood, Trade Republic, eToro etc.): eliminates explicit brokerage fees, massively lowering the barrier to retail equity trading
- **But the cost does not disappear** — it is shifted:
 - **Payment for order flow (PFOF)**: the broker routes your market orders to a wholesaler (market maker) who pays the broker for the right to execute against them
 - The wholesaler profits from the spread and from trading against order flow that is predictable and uninformed
 - The broker earns revenue without charging you; you may receive slightly inferior execution prices
- **Scale**: Robinhood's popularity forced Schwab, Fidelity and TD Ameritrade to eliminate commissions in 2019; US brokerages earned \$3.8bn in PFOF revenue in 2021 alone; Citadel Securities handles >80% of US retail equity orders
- **Regulatory divergence**: UK banned PFOF in 2012; EU banned it from March 2024; the US still permits it — the SEC's reform proposals were withdrawn in 2025

GameStop: January 2021 – the story

The most significant retail market structure event in a decade.

The mechanics:

- GameStop (GME) was a heavily shorted stock – approximately **140% of float** sold short
- Reddit's r/WallStreetBets community coordinated aggressive buying to trigger a short squeeze
- GME rose from \$17 in early January to a pre-market high of **over \$500** on January 28
- January 27, 2021: US equity and options volume hit all-time highs (24.5bn shares; 57.1mn contracts)
- Melvin Capital (short GME) lost 53% of AUM (\$6.8bn); required a \$2.75bn emergency bail-out from Citadel and Point72

The halt:

- Pre-open January 28, Robinhood faced a **\$3.7bn margin call** from the DTCC clearinghouse
- Robinhood had only \$700mn in collateral – it was short \$3bn with hours to find it
- At 10am Robinhood **prevented buying** of GME and other meme stocks
- GME fell from \$500 to \$193 intraday
- Congressional hearings followed; Robinhood CEO testified before the House Financial Services Committee

GameStop: microstructure lessons

The episode simultaneously illuminates almost every concept from this module:

- **Short selling and market efficiency:** does coordinated retail buying constitute manipulation, or is it legitimate informed trading against an overvalued short position?
- **PFOF and routing:** Citadel Securities was simultaneously (a) Robinhood's largest PFOF revenue source and (b) one of the firms that bailed out Melvin Capital – creating an apparent (though contested) conflict of interest
- **Liquidity under stress:** meme-stock volatility reduced market-maker willingness to quote – bid-ask spreads widened dramatically even for large-caps as contagion spread
- **Regulatory response:** US moved to T+1 settlement (May 2024); PFOF reform proposed but withdrawn; enhanced short-sale disclosure rules under consideration

Beyond equities: OTC dealer markets

Everything we have studied so far focuses on equity markets with CLOBs. Much of the world's financial market activity takes place in a very different structure.

The OTC dealer model:

- No central exchange — bilateral trading between dealers and customers
- **Dealers quote prices** to customers on request; customers shop among dealers or use a broker
- This is the **dealership structure** from lecture 1, but without an exchange enforcing any transparency regime
- No *pre-trade* transparency: you cannot see a consolidated order book
- Post-trade transparency has improved but varies enormously by asset class
- **Markets that use this structure:**
 - FX spot and derivatives (\$7.5 trillion/day globally)
 - Government bonds (Treasuries, Gilts, Bunds) and corporate bonds
 - OTC derivatives (swaps, options)

FX market structure

The world's largest financial market: average daily turnover of \$7.5 trillion (BIS 2022)

Two-tier structure:

Interdealer market

- **EBS** (now CME-owned) and **Refinitiv/Reuters Matching** are the “primary markets”
- Effectively CLOBs for dealers only – the main source of price discovery
- Interdealer volume on primary venues has *declined* significantly since 2010

Dealer-to-customer market

- **Multi-dealer platforms** (e.g. Cboe FX, FXall): customers request quotes from multiple dealers
- **Single-dealer platforms:** a bank's proprietary streaming price
- **Internalisation:** large banks increasingly match flows internally

Parallel with equities: the FX market has fragmented dramatically since 2000 – c. 60% of trading is now electronic (vs. <30% in 1998), primary venues have lost market share to single-dealer platforms and internalisation. [▶ Story](#)

Stress in OTC markets [1]

The same asymmetric information and inventory-risk channels from L2 operate in OTC markets – but the institutional setting creates additional amplification mechanisms.

March 2020 – US Treasury market:

- US Treasuries are the world's benchmark “safe-haven” asset – normally the most liquid securities in existence
- In the week of 9–13 March 2020, bid-ask spreads on 10-year Treasuries widened to **ten times** their normal level
- Primary dealers – banks *obligated* to make markets in Treasuries – hit balance-sheet limits: they could not absorb selling pressure
- The *inventory* channel was dominant: $\sigma^2 \uparrow$ dramatically, $\rho\sigma^2 \uparrow$ even faster
- The Federal Reserve ultimately purchased **\$1.6 trillion** of Treasuries in two weeks to restore market function

Stress in OTC markets [2]

Why was this worse than equity-market stress?

- No CCP: dealers cannot net positions across counterparties as easily as in equity markets
- No HFT market makers to step in when humans withdraw
- Balance-sheet constraints bind more tightly in OTC markets (Basel III capital requirements)
- The market literally depends on a small number of dealer banks being willing to quote

Part 1: Summary

- We now have a clear idea of how to **measure liquidity**: spreads, depth and resilience
- We understand how liquidity is **determined**: asymmetric information and dealer risk aversion (including how both channels amplify in stress)
- We have explored **modern market structure**:
 - Fragmentation, dark pools and the regulatory response (DVC → SVC in the EU; caps removed in the UK)
 - HFT: broadly positive for average liquidity but arms-race costs are quantitatively significant (Aquilina, Budish and O'Neill, 2022)
 - The Citigroup 2022 incident shows that algorithmic control failures remain a serious systemic risk
- We have seen how the same forces play out in **retail markets** (GameStop 2021; PFOF and the zero-commission model)
- And how equity microstructure concepts extend to **FX and bond markets**: same theory, different institutions – OTC structure, last look, balance-sheet constraints, and the role of central banks as market makers of last resort